MAGUS: an LLM Multi-Agent Tool for Generative Accessibility of GUIs from User Stories

Recent breakthroughs in Generative Artificial Intelligence (AI) have enabled the automatic creation of Graphical User Interfaces (GUIs). However, accessibility is still often treated as a post hoc activity, resulting in inconsistent and reactive integration. In this context, we present MAGUS, an LLM Multi-Agent tool for generating accessible GUIs from User Stories. (A) A User Story Generation Agent transforms generic user stories provided by the UX designer into accessibility-oriented variants aligned with WCAG principles[1]. (B) Component Selection Agent maps these accessibility-oriented stories to appropriate Material Design (MD) components, while a Designer Agent generates an intermediate GUI script representation that embeds accessibility attributes such as ARIA roles and labels. A Renderer Agent and a Converter Agent are responsible for generating prototyping and GUIs based on the AI Figma make[2]. (C) Accessibility Evaluation: a set of Specialized Accessibility Agents evaluates the generated GUI script across the four WCAG dimensions (Perceivable, Operable, Understandable, Robust). Each agent category has supportive agents, which are created as needed based on accessibility issues to improve accessibility retention. Then, they produce structured accessibility reports, which are fed back into the pipeline to enable iterative refinement. Through preliminary expert assessment, MAGUS effectively creates GUIs that reflect functional intent while meeting WCAG accessibility standards. These early findings introduce MAGUS as a potential tool to contribute to generative accessibility UI while maintaining a Human-in-the-Loop approach, in which designers supervise and refine generative outputs rather than manually designing interfaces.

CCS CONCEPTS • Generative UI; Accessibility; Human-AI collaboration; Human-in-Loop, Multi-Agent Systems

1 INTRODUCTION

Generative AI is quickly changing the landscape of user interface design, making the process more intuitive and efficient. To guide this process, GUI generation is often based on User Stories, which UX designers use to maintain focus on user needs [1], [2]. User stories are written in natural language and can be leveraged with Large Language Models (LLMs) to translate functional requirements into structured interface representations [1], 2]. Advanced LLM-powered approaches have the potential to bridge the gap between user-driven requirements and tangible interface designs, particularly when user stories are used as foundational semantic artifacts for prototyping [3], [4] [5].[6] However, recent studies have shown that LLMs often exhibit hallucinations, conflicting outputs, failures in causal or mathematical reasoning, uncertainty, or overconfidence [7]. One solution is to distribute workloads among multiple targeted LLM agents that collaborate to deliver more reliable outcomes [8]. While multi-agent systems have the potential to enhance the interface generation process, thereby reducing human effort while meeting user needs, not all users benefit, as accessibility considerations are often added as an afterthought following the completion of interface design [9], [10]. [11], [12], [13] . This results in inconsistent application of accessibility features or even their oversight [10] [14]. In this context, we propose MAGUS, an LLM-based multi-agent tool that presents a new perspective, Generative Accessibility, which embeds accessibility considerations at every stage of the interface creation process, ensuring that inclusiveness is woven into the design from the outset. To situate MAGUS within the broader context of generative and adaptive interfaces, we formulate the following research questions to inform our preliminary results and discussion: **RQ1.** How effective is our tool in generating accurate and reliable accessibility user stories that address the needs of people with disabilities? **RQ2.** To what extent can MAGUS-generated GUIs translate the accessibility requirements captured in user stories? and **RQ3.** How does the MAGUS affect users' perceived usability, trust, and cognitive workload compared to the baseline?

2 PRELIMINARY EVALUATION

2.1 Evaluation Process

We evaluated MAGUS by recruiting 12 experts via Prolific[3] [15] to rate the user stories and the generated GUIs based on fidelity, accuracy, and clarity using Likert scales and questionnaires. Metrics such as precision, recall, and F1 score were computed for comparison with the ChatGPT-5 baseline. Additionally, 28 computer science students tested MAGUS and the baseline, completing tasks and usability questionnaires.

[1] https://www.levelaccess.com/resources/must-have-wcag-2-1-checklist/
[2] https://www.figma.com/make/
[3] Prolific Academic Ltd. 2025. Prolific. www.prolific.com.

2.2 Preliminary Findings

RQ1: Experts evaluated each MAGUS-generated user story for functional fidelity, accessibility accuracy, clarity, and actionability, where initial evaluation revealed that of 624 stories, 38 (~6.1%) were excluded for irrelevance, redundancy, or unclear intent, leaving 586 for analysis. The exploratory results indicate that expert ratings were positive, with average and median scores addressing accessibility needs (7.8) and clarity/usefulness (8). However, the score for maintaining a functional intent (6,5) is relatively low; in some cases, the functional requirement in the original user story does not fully appear in the generated accessibility user story. **RQ2:** Experts' initial evaluations of the generated accessible GUIs were positive, with MAGUS's performance reliably translating the functional goals of accessibility-focused user stories into working GUI elements, with a Functional Accuracy Index averaging 8.12 out of 9. The layouts keep elements visually grouped and aligned. **RQ3:** The preliminary results revealed highly positive perceptions across all aspects, showing that the MAGUS tool was both effective and easy to use when interacting with the baseline, which they did not recommend because they encountered irrelevant GUI components and confusing layouts.

3 IMPLICATIONS FOR HUMAN-COMPUTER INTERACTION PRACTICE

Generative accessibility is reshaping how designers create accessible interfaces. Instead of treating accessibility as a post hoc check, it is now integral to interface design from the outset. This shift means designers are less focused on manually adding accessibility features and more on guiding and reviewing the outputs of generative tools. With this approach, accessibility is woven into the design, helping ensure it is consistently addressed and reducing the need for expensive fixes. Nevertheless, as generative accessibility becomes standard in HCI practice, new hurdles emerge. The explainability and transparency of how generative AI makes decisions are unclear. What forms of transparency are needed to make generative accessibility decisions understandable and trustworthy? Moreover, building trust in these tools means keeping humans involved, so oversight and guidance remain central to the process. How can designers oversee and verify accessibility-focused generative systems in a way that enhances rather than diminishes the benefits of generative AI, even when they can manually create accessible interfaces? This might pave the way for a new form of Human-in-the-Loop, with a shift in designers' roles from creating UIs to supervising and validating generative AI outputs, enabling solutions that produce interfaces that are not only functionally dependable but also inherently inclusive by design.

4 LIMITATIONS AND FUTURE WORK

This work is exploratory and reflects a work in progress. The preliminary results offer a glimpse of what the tool can achieve, but they are not yet final. To gain a clearer understanding and greater confidence in its effectiveness, the tool still needs further development and more extensive testing. The MAGUS tool primarily relies on a limited dataset of 208 user stories, which could affect subsequent tasks and reduce the efficiency of outputs. In fact, some generated stories may be superficial or fail to encompass the full spectrum of disability experiences. The next step in this study is to broaden the dataset by incorporating more detailed, domain-specific accessibility stories and involving actual users with disabilities, thereby improving contextual precision and inclusivity. On the other hand, MAGUS evaluation is conducted in highly controlled environments with a relatively small group of accessibility experts and participants. More extensive real-world studies are necessary to better reflect the diverse nature of design processes, domains, and cultural perspectives on accessibility in GUIs.

5 CONCLUSION

This work introduces MAGUS, a novel LLM-based multi-agent tool that bridges the gap between accessibility user stories and GUI generation and presents the concept of Generative Accessibility, which weaves accessibility reasoning directly within generative UI workflows. In a preliminary study, early results indicate that MAGUS can generate accessible user stories and GUIs that meet accessibility standards; however, they also highlight the need for stronger validation and broader accessibility evaluation. To move forward with generative accessible interfaces, it is crucial to focus on human–AI collaboration, transparency, and reliable, accessibility-conscious generation. We can develop digital experiences that are truly inclusive and reach a broader audience, making accessibility a fundamental part of the design process rather than an afterthought.

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